2021 International Conference on Software Engineering & Computer Systems and 4th International Conference on Computational Science and Information Management (ICSECS-ICOCSIM)

2021 International Conference on Software Engineering & Computer Systems and 4th International Conference on Computational Science and Information Management (ICSECS-ICOCSIM) | 978-1-6654-1407-4/21/$31.00 ©2021 IEEE | DOI: 10.1109/ICSECS52883.2021.00114

Alarming Assistive Technology: An IoT enabled Sitting Posture Monitoring System

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***Abstract*— The usage of smart devices is tremendously booming up in recent years. The users of smartphones are increasing over the years as well. However, there is a great tendency that users gradually change their posture while using smart devices over long hours. This might potentially lead to several health risks such as damaged spinal cord or back pain on the users. In this paper, we propose a monitoring system with an IoT framework that could monitor the sitting posture of the user while using the smart devices. This proposed alerting system integrates the sensors and wireless networks for data accumulation for monitoring a person's sitting posture and alert them to correct their sitting posture.**

Keywords—Internet of Things, posture monitoring, sitting posture, sensor, wireless network

I. Introduction

The advanced technologies and entertaining features of the smart devices makes the users spend an enormous amount of time with it [1]. Being attracted with those smart features, there is a great tendency among the smart device users to spend a long time with it. As a result, the users get several health risks such as neck pain, back pain, bad postures, shoulder strain etc.

The usage of smartphone users is gradually escalating worldwide. This surpasses nearly three billion people and is forecast to increase further in the next few years. According to the recent statistics on worldwide smartphone users, many countries are having the greatest number of smartphone users. This shows a massive amount of increase in the usage of smartphones in recent years [2]. Together, the number of computer users has also increased, because computers are more convenient and reliable than the traditional ways of doing the tasks [3]. However, this increase in smart device usage reflects the potential growth in the health risks of the smart device users.

Worldwide, the impact on the economic part of the health issue plays a vital role. This is because back pain or spinal injury is one of the most common reasons for every human to absent themselves from their work [4]. Recent studies show that almost 7 % of the world population could be affected by back pain at any time. This would worsen the earning capability of the people despite their age [5].

Therefore, the need to assess the sitting posture is clearly identified. Hence, this paper develops a monitoring system to assist the users to have a better sitting posture while using the smart devices.

The posture of the human body can be detected using two broad categories namely, invasive and non-invasive methods. The invasive method of detection is by using the equipment on the human body such as sensors. The non-invasive method is by using any devices such as cameras to collect the posture details and then process the data to classify the posture. Moreover, studies proved that invasive methods provide better accuracy in posture detection [6]. This would help the researchers to develop a higly practical design to reduce the bad sitting posture of smart device users [7].

An IoT infrastructure integrates the sensors and other devices through the wireless networks such as Bluetooth, Wi-Fi, etc) [8]. Being the invasive method of posture detection, the sensors are used in IoT structure to accumulate the data from the user. The collected data will be analyzed to detect the correct posture [9].

Another posture detection solution is discussed that uses wearable posture recognition with wearable sensors. Here, the sensors are integrated with a microelectromechanical system [10]. The authors collected the data of human body posture from the sensors. There are several sensors used in the IoT infrastructure such as accelerator sensor [11], velocity sensor and pressure sensor. Posture of the human body is related to walking, running, sitting and falling, etc [12]. Most people tend to have a bad posture during prolonged use of smart devices [13].

This paper presents the enhancement of our previous work [14] which investigated the basic conceptual design of the monitoring system. Therefore, this work has two main contributions:

- Implement the IoT infrastructure on a sitting chair that helps to accumulate data, process and visualize on smart devices such as smart-phones, computers, etc.
- Develop the mobile application for users to visualize the data collection from the sensor and assess the collected data from a mobile application which is linked via Bluetooth to the sitting chair.

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DOI 10.1109/ICSECS52883.2021.00114

II. System Architecture and Design

The proposed IoT based system is focused to observe the sitting posture of the human body while using smart devices.

A. System Architecture

Fig. 1 illustrates the architecture of the system consisting of ultrasonic sensors, microcontroller, rechargeable battery and Bluetooth module. The sensors are utilized for the purpose of collecting data from the user sitting on the chair.

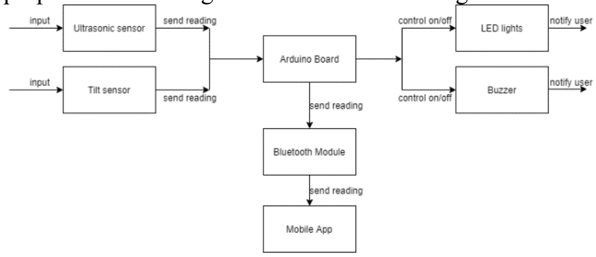


Fig. 1. Architecture of the proposed monitoring system

According to the proposed design, the data of the sitting posture is accumulated by using ultrasonic and tilt sensors. The Ultrasonic sensor collects data of the user's upper torso. The tilt sensor measures the data of the user's neck posture. The accumulated data is then transferred to the microcontroller to make decisions on collected data. The sensor data can be visualized through a mobile application. The data is transferred to a mobile application via Bluetooth. Accordingly, the bad posture of the user is identified and the user is alerted using buzzer and LED lights. This alerting system will enable the users to correct their sitting posture.

B. Circuit design

The circuit design of the system is shown in Fig.2.

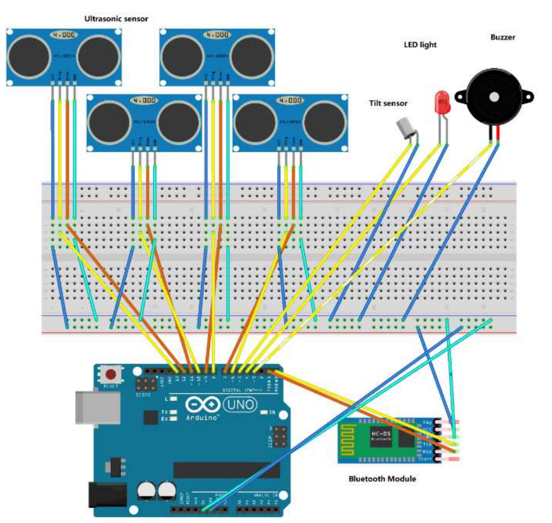


Fig. 2. Circuit design of proposed system

The circuit and corresponding wiring system to interconnect all the components are shown in Fig.2. The wires in orange and yellow colours are used for input/output purposes of the system. The wire labelled with blue colour denotes the voltage supply. The wire labelled with cyan denotes grounding.

C. Implementation of sensors on sitting chair

The sensors of the IoT infrastructure are implemented on a sitting chair and headphone. This is demonstrated in Fig.3 and Fig.4. The rechargeable batteries, Arduino system and Bluetooth module are placed separately. The Ultrasonic sensors are attached on the backseat of the chair that is clearly depicted in Fig.4. The Tilt sensor is equipped on top of the headphones. The LED light is installed in front of the user so that the user can visualize the LED light.

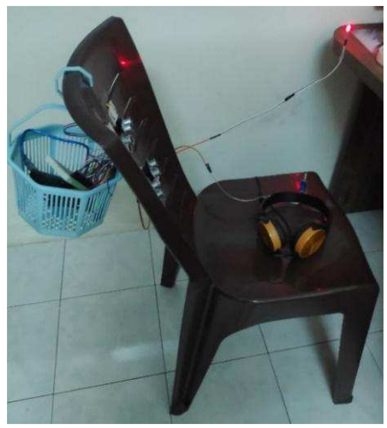


Fig. 3. Sensors setup on sitting chair and headphones

In Fig.4, the four ultrasonic sensors are installed at the backseat of the sitting chair. The four labels are Top Left (TL), Top Right (TR), Bottom Left (BL) and Bottom Right (BR) part of the user's upper torso.

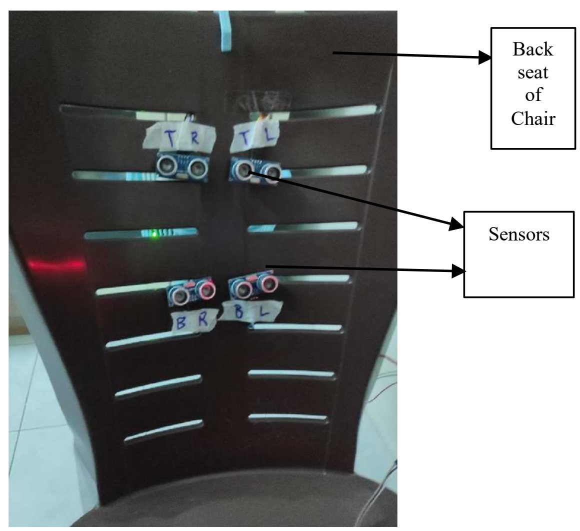


Fig. 4. Implementation of sensors on sitting chair

D. Workflow of the proposed system

The flow of the overall working of the proposed system is illustrated in Fig.5. Once the user sits on the chair and when the power is switched on, the sensors start to collect data on the sitting posture of the user.

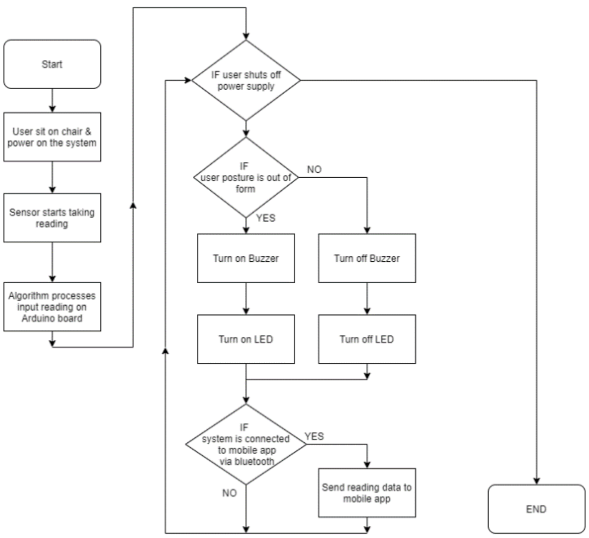


Fig. 5. Workflow of the proposed system

The collected data is transferred to the microcontroller and it is programmed on good and bad posture of human sitting posture while using the smart devices. The programmed microcontroller will determine whether the user is sitting in bad or good posture. If the user is sitting in bad posture, the LED light and buzzer will be turned on to alert the user. After the alert, if the user corrects the sitting to a good posture, the LED lights and buzzer will be turned off. If the user connects the system to a mobile application via Bluetooth, the data collected from sensors is transferred to the mobile application for the user to monitor their sitting posture data in real time.

III. THE EXPERIMENTAL PROTOCOL OF THE PROPOSED SYSTEM

The experiment of the proposed system is conducted in an office environment with the users involving prolonged usage of computers and smartphones. The users are asked to sit on the chair in their normal posture so that the user's feet are in contact with the ground.

A. The Posture Detection Algorithm

The sitting posture works as discussed below. The detection algorithm is illustrated in Fig.6. Firstly, when the system is powered on, all variables will be initialized and assigned with value 0. The system starts to collect data from ultrasonic sensors and tilt sensors. The process is continued until the system is powered off. There are four ultrasonic sensors used in this system, each responsible to collect the distance between the user's upper torso and the chair. Each ultrasonic sensor is placed at TL, TR, BL and BR of the chair to collect more accurate reading of user's upper torso.

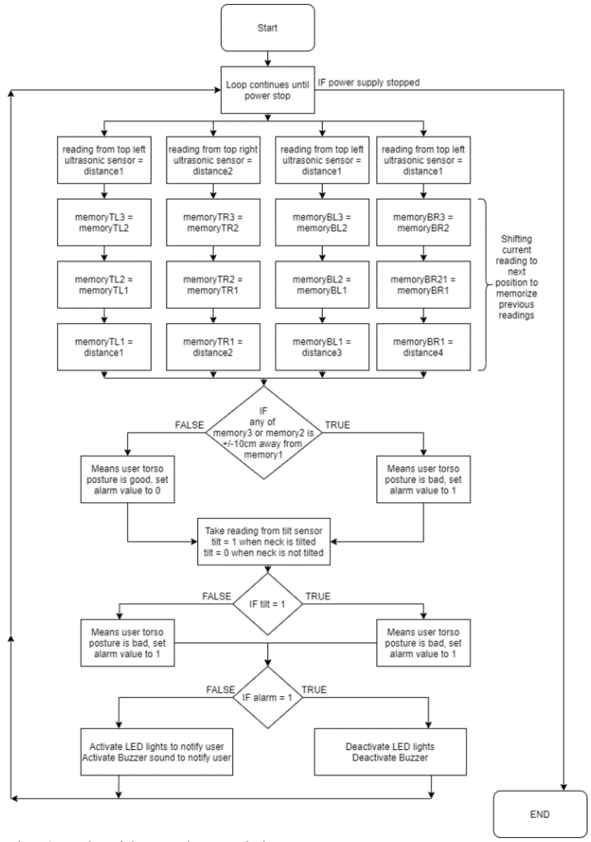


Fig. 6. Algorithm to detect Sitting Posture

The data of the TL ultrasonic sensor that is placed at the top left of the chair is recorded in variable distance1. The data of the TR ultrasonic sensor installed at top right of the chair is recorded into distance2. The data of the BL ultrasonic sensor installed at bottom left of the chair is recorded into variable distance3. The data of the BR ultrasonic sensor installed at bottom right of the chair is recorded into distance4.

Then, the memory value of TL1 is shifted to TL2, and TL2 is shifted to TL3, and the new data from distance1 (TL ultrasonic sensor) is recorded into memory TL1. The values of TL2 and TL3 are compared to TL1. If either of the values of memory in TL2 or TL3 is 10 (10cm) less or more than TL1, the value of the variable alarm will be assigned to value '1'.

The comparison between previous and current data of the ultrasonic sensor is executed for other ultrasonic sensors installed at top right, bottom right and bottom left of the chair, by comparing memoryTR, memoryBL and memoryBR.

Secondly, the posture of the neck of the user is collected by using a tilt sensor. If the tilt sensor detects the asymmetric user's neck posture, the variable tilt will be assigned with value '1'. When the value of tilt is assigned with value '1', the value of variable alarm will be assigned with value '1'.
When the value of the variable alarm is assigned with value '1', the microcontroller sends an alarm through the LED lights and buzzer to notify the user. In order to switch off the LED lights and buzzer, the user must sit in good posture for an extended period of time to reassign the value of the alarm to '0'.

B. Determination of Upper Torso Posture detection

The sitting position of the user's upper torso is determined by comparing the current data with previous data in the variables. This is demonstrated in Fig.7. If the current data and two previous data are identical, this means that the user is sitting in a fixed posture. However, when current data and either of any one of the two previous data are non-identical, this means that the user's sitting posture is deformed over time, and the user is probably sitting at bad posture.

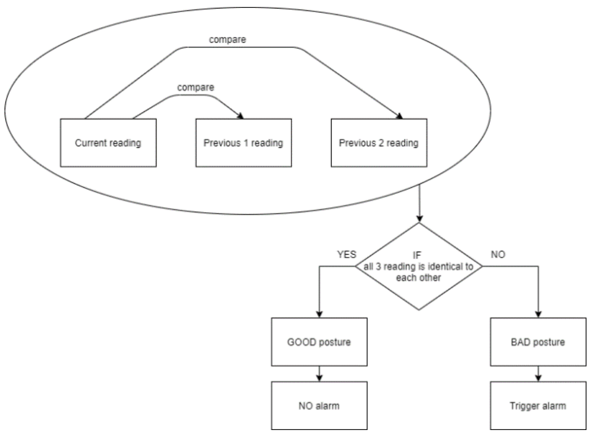


Fig. 7. Determining Sitting Posture

The reason behind this data comparison is, when a user prolongs his/her sitting posture and moves slightly, it is still not determinant enough to say that the user is sitting in a bad posture. This is because the user's sitting posture might be moved slightly and not detectable by the system. Hence, the comparison is made between current and two previous data of the sensors. Even if the user is sitting for a long time and moves slightly, the current data of posture is still compared with data of previous data from a long time ago, effectively increasing the accuracy of posture detection.

Additionally, when the user's upper torso is in contact directly with an ultrasonic sensor, this implies that the user is sitting in bad posture, hence this triggers the alarm. When the user's upper torso is in direct contact with an ultrasonic sensor, the sensor will not be able to measure distance properly. This is because, the soundwave emitted is partially absorbed by user's body and clothes, the reflected soundwave is no longer acts as a determinant for distance measurement

C. Determination of Neck Posture detection

The detection of the user's neck position is determined by using Tilt sensor. The tilt sensor is also referred to as a rolling ball sensor or tilt switch which is to detect the inclination. The tilt sensor is cylindrical and it contains a free conductive rolling ball inside with two conductive elements beneath. The two conductive elements are not connected directly. When an object is tilted, the ball touches both conductive elements and acts as a bridge for the current to pass through, indicating an object is tilted through the presence of electrical current.

The tilt sensor is attached to a headphone. The tilt sensor is only activated when the user is facing downwards while wearing the headphone. When the user is facing downward for a long time, this could damage the health of the user's neck. Hence, when the user is facing downward, the system will trigger an alarm using LED light and buzzer to notify the user is in bad posture, and the user should reposition himself/herself in a good posture.

D. Mobile Application Development

The mobile application is developed according to the flow shown in Fig.8. The mobile application starts working if the Bluetooth module of the mobile is turned on. Once the Bluetooth is turned on, the user should connect the application to the system. After the successful connection, the user data from sensors are collected and displayed in the mobile application to inform the user about monitoring status.

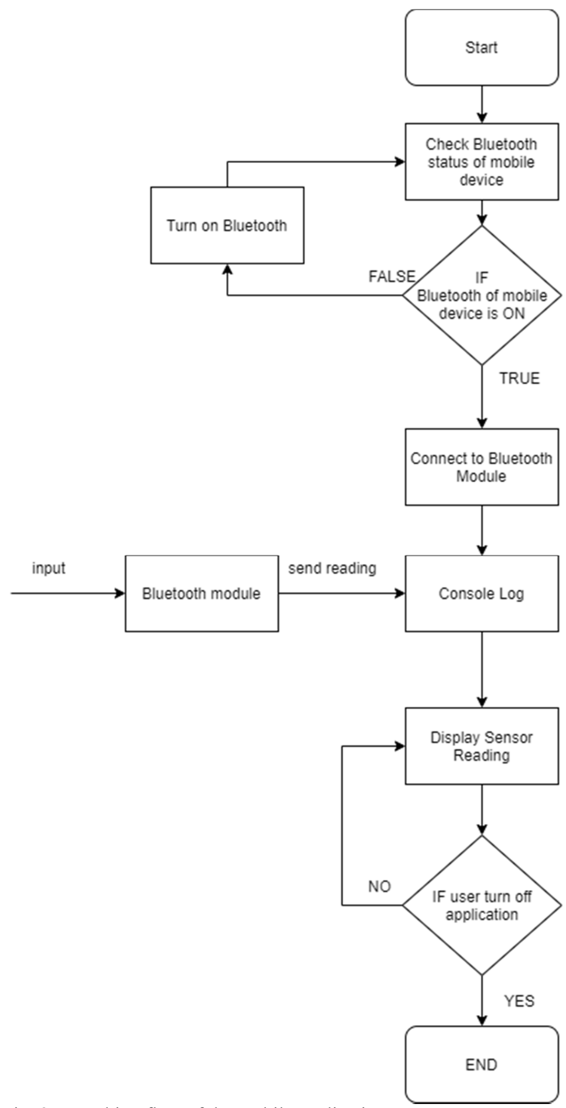


Fig. 8. Working flow of the mobile application

IV. Results and Discussion

The performance of the proposed system is shown by the following results under several conditions. Fig.9 shows the results of the experiment conducted with the minimum required distance between user body and ultrasonic sensor.

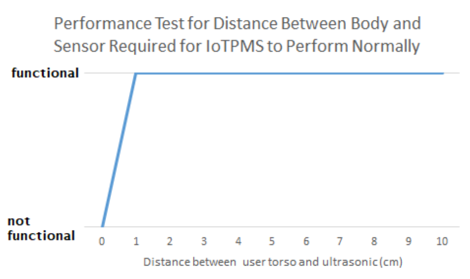


Fig. 9. Performance results: Distance between human Body and Sensor in Normal condition

When the user's upper torso is in direct contact with an ultrasonic sensor, the sensor cannot collect accurate data because the emitted sound waves are partially absorbed by the user's body and clothes. The results show that the system requires minimum of 1 cm in between ultrasonic sensor and user's body in order for the proposed system to be functional.

Fig.10 shows the duration of the proposed system that can work continuously without failure. It shows that at 180th minute, the system becomes non-functional. The reason for not being functional is that the rechargeable battery ran out of power.

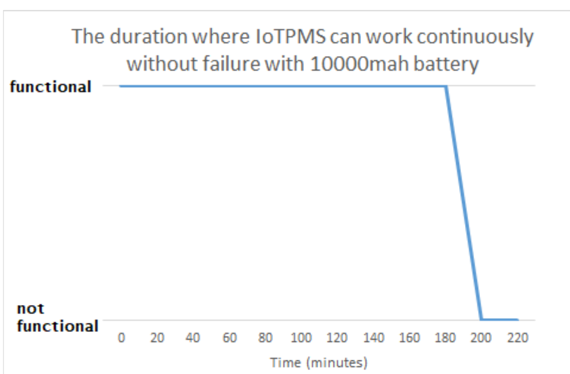


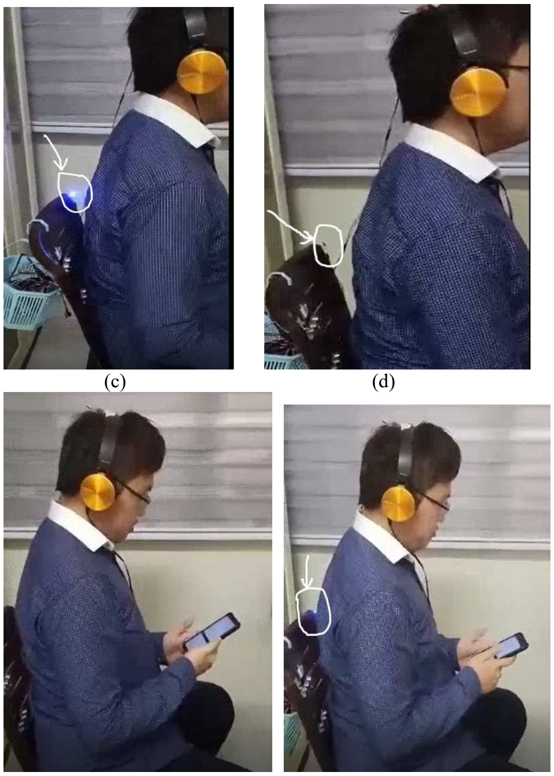
Fig. 10. Performance results: System's Working duration without failure

The mobile application interface in Fig.11 shows the sensor data received via Bluetooth module and displayed on the mobile device. The three data of the latest distance between the user's torso and the ultrasonic sensors on TL, TR, BL and BR of the sitting chair that is collected in centimetres (cm) is also visualized in the mobile application.

Fig. 11. Mobile application interface

The user sitting in his/her normal posture is shown in Fig.12(a). This posture is considered as normal for this study after consulting a physiotherapist. The user sits by bending his backbone which is shown in Fig.12(b). This sitting posture data is collected and this triggers the alarm to the user through the LED light and buzzer sound. The alarming LED light is shown in Fig.12(c). By using these alerting signals, the user can correct their position. Once the posture is corrected, then alerting signals will be turned off automatically as shown in Fig.12(d). The posture of the neck is also monitored. The user using his smart phone by lowering his neck is shown in Fig.12(e). When this position is maintained for 5-10 minutes, the alerting signal will be arised through LED light and buzzer signal. Accordingly, the user should correct the neck posture to turn off the alarming signals. The proposed IoT enabled monitoring system is aimed to help the smart device users to sit in a good posture if they intend to sit for prolonged time. Monitoring the sitting posture of humans helps to maintain correct body posture and reduce several health problems.

(a) (b)



(c) (d) (e) (f)

Fig. 12. Views of sitting posture (a) normal (b) bad posture (c) alarm signal ON (d) alarm signal OFF (e) Neck posture (f) alarm signal ON

V. CONCLUSION

This paper proposed an IoT enabled monitoring system to monitor the sitting posture and alert the user to correct their posture. The system is designed with integrated IoT infrastructure. The monitoring system can accumulate posture data by focusing the study on the neck and the backbone of the user. This research work is still in the preliminary development and the work we have presented here shows that the proposed system can perform with accurate results. More research is required to quantify the body sitting posture and analyze the data using machine learning algorithms to enhance the reliability and validate the performance of the system.

Acknowledgements

This work was supported by Universiti Malaysia Pahang, Malaysia under a research grant, RDU1803123.

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